

Umma Marium Alam

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About

Final-year CSE student at BRAC University (CGPA: 3.71/4.00) with production-grade web development experience, state-of-the-art ML thesis research, and peer mentoring. Currently deepening ML research through GCI World 2026 April program (Matsuo Lab, University of Tokyo). I build and research.

Skills

Web & App Dev

React, Laravel, MERN, PHP

ML & AI Research

Diffusion, CV, Scikit-learn

Tech Stack: **React** • **Node.js** • **Laravel** • **Python** • **TensorFlow** • MongoDB • MySQL

Research

Defended Thesis (2025)

Fast-DDIM VTON: Multi-View Diffusion-Based Virtual Try-On

Proposed a novel framework combining Adaptive Cross-Pose Attention, Partial Multi-View Learning, Pose-Adaptive Noise Scheduling, Region-Aware Diffusion, and accelerated DDIM sampling. First system for culturally specific garments—introduced the BTD dataset.

SSIM 0.8834 ↑ LPIPS 0.0987 ↓ FID 3.8465 ↓ KID 4.1375 ↓

ML Projects

Stroke Risk Prediction

2025

Python, Scikit-learn, TensorFlow • 4,000+ patients

- Logistic Regression: **95.2% accuracy** / **AUC 0.842**
- Analysed class imbalance trade-offs across Accuracy, F1, and AUC-ROC.

California Housing Price Prediction

2025

Python, Scikit-learn, Jupyter

- Linear R^2 0.654 → Random Forest R^2 **0.767** (+17%)
- Feature engineering and Grid Search hyperparameter tuning.

Web Projects

Notes App (MERN)

2025

React, Node.js, Express, MongoDB

- Full-stack CRUD with React frontend and Express REST API.
- Component-based architecture with real-time state updates.

Handy Laundry

2025

Laravel, PHP, MySQL

- End-to-end booking system with admin dashboard.
- Role-based access control (customer/staff/admin).

Education

B.Sc. Computer Science & Engineering

BRAC University, Dhaka

CGPA: 3.71/4.00

Sept 2022 – Present

- Coursework: ML, Computer Vision, Data Structures, Algorithms.

HSC & SSC (Science)

BNMPC

GPA: 5.00/5.00

2019 – 2021

Leadership

- **Senior Executive, BRACU Computer Club** (2024-2026): Mentored junior members in programming and project work.
- **Director, Peace Café BRACU** (2024-2026): Led team operations and event coordination.